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Daily Use of a Broad-Spectrum Cannabidiol Supplement Produces Detectable Concentrations of Cannabinoids in Urine Prohibited by the World Anti-Doping Agency: An Effect Amplified by Exercise

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ABSTRACT

Background: Cannabidiol (CBD), a non-intoxicating phytocannabinoid, is used by athletes to enhance recovery and manage other conditions (e.g., poor sleep, anxiety). Although CBD is not prohibited by the World Anti-Doping Agency (WADA), other cannabinoids found in "broad-spectrum" CBD products (e.g., cannabigerol (CBG), cannabidivarin (CBDV)), remain prohibited. **Objectives:** This study aimed to determine whether 10-weeks' use of a broad-spectrum CBD product (150 mg·day [containing trace concentrations of CBG]) could lead to detectable concentrations of prohibited cannabinoids in urine and plasma. The influence of moderate-intensity exercise was also assessed. **Methods:** Thirty-six healthy individuals (47% male) self-administered either a broad-spectrum CBD product (n = 31; CBD) or a visually identical placebo (n=5; PLA) for 10-weeks. After 10-weeks, participants completed a fasted, 90-minute bout of moderate-intensity exercise (55% VO_{2peak}). Blood and urine samples were collected at baseline (pre-supplementation) and pre- and post-exercise. **Results:** No cannabinoids or metabolites were detected at baseline in either the PLA or CBD group. Following 10-weeks of supplementation, urinary concentrations of CBD and its metabolites (6-OH-CBD, 7-COOH-CBD, 7-OH-CBD) were present. CBG and CBDV were also detected in 42% and 68% of pre-exercise samples, respectively. Urinary concentrations of 6-OH-CBD (P=0.006), 7-OH-CBD (P=0.009), CBD (P=0.043), CBG (P=0.0023) and CBDV (P=0.033) also increased from pre- to post-exercise. CBG and CBDV were detected in 74% and 84% of post-exercise samples, respectively. Concentrations of Δ^9-THC or its metabolites (11-OH-THC, 11-COOH-THC) were not present at any timepoint. **Conclusions:** Daily use of a broad-spectrum CBD supplement resulted in detectable urinary concentrations of WADA-prohibited cannabinoids in urine. Exercise appeared to increase concentrations of these cannabinoids. Therefore, athletes should avoid consuming broad-spectrum

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CBD products, given the potential associated anti-doping risks. **Key Words:** Δ^9-THC, CANNABIGEROL, CANNABIDIVARIN, CANNABINOID DETECTION, ATHLETE SUPPLEMENTATION

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INTRODUCTION

Cannabis sativa L. is an annual herbaceous plant that has been used for millennia for a variety of purposes (1). The cannabis plant possesses many bioactive molecules, most notably >140 terpenophenolic compounds known as phytocannabinoids. The best-studied of these is Δ^9-tetrahydrocannabinol (Δ^9-THC), the main intoxicating component of cannabis, while cannabidiol (CBD), cannabigerol (CBG) and cannabidivarin (CBDV) are non-intoxicating cannabinoids with emerging therapeutic potentials (2, 3). That said, there have been conflicting reports regarding the utility of cannabinoids, specifically CBD in supporting sports performance and recovery (Gillham et al., 2024). These formulations can either be "isolate" products, where CBD is the only cannabinoid present, "full spectrum" products, where CBD, Δ^9-THC and other phytocannabinoids (e.g. CBG, CBDV) are present, or "broad-spectrum" products, where CBD, other phytocannabinoids, but not Δ^9-THC, are present (4).

In recent years, interest and supplementation with commercially available CBD has become prevalent in professional sport (5, 6). For example, 26% of professional rugby players surveyed (and 39% of those aged >28 years) reported either currently using or having previously used CBD (7). The most common reasons for use were to enhance recovery (80%), improve sleep (78%), reduce anxiety (32%), and treat 'other' medical conditions (e.g., concussion). Most (67%) of respondednts also reported a perceived benefit from supplementation (7). Indeed, some early-stage evidence suggests that CBD has anti-inflammatory, neuroprotective, analgesic and anxiolytic effects (8).

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An important consideration, however, surrounding athlete use of CBD are the anti-doping regulations pertaining to cannabinoids (9). The World Anti-Doping Agency (WADA) Code prohibits all cannabinoids other than CBD in any concentration, except for Δ^9-THC, where urinary concentrations of Δ^9-THC metabolites (11-hydroxy-tetrahydrocannabinol [11-OH-THC] and 11-carboxy-tetrahydrocannabinol [11-COOH-THC]) should remain below a 150 ng/mL threshold (8, 9). The trace concentrations of 'other' phytocannabinoids (e.g., CBG, CBDV) present in many full spectrum and broad spectrum CBD products, therefore, constitute a significant concern given that any concentration is theoretically an anti-doping violation.

An additional concern is that the cannabinoid concentrations within commercially available products often differ substantially from those stated on the labels. For example, in Mississippi, only 12% of products tested contained cannabinoids that were within 20% of the label claims (Gurley et al., 2020). Moreover, some studies have also identified the presence of Δ^9-THC in "Δ^9-THC-free" formulations (Johnson et al., 2022). Similar findings have also been elucidated in the United Kingdom, with one study reporting only 38% of products tested were within 10% of the advertised CBD content (Liebling et al., 2020). Another recent investigation observed that only 8% of products tested had quantifiable concentrations of CBD within 10% of those advertised (Johnson et al., 2023). Subsequently, the general message from sport nutritionists is that athletes should a) remain cautious when supplementing with CBD, or b) avoid CBD supplements. However, athletes often consume these products without the advice or knowledge of qualified personnel (7).

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Finally, it is important to recognise that the pharmacokinetics of CBD, Δ^9-THC and other phytocannabinoids are notoriously erratic and appear to be influenced by several factors (e.g., genetics, drug interactions) in addition to diet and exercise (10). For example, some evidence suggests that the highly lipophilic nature of cannabinoids means they can be sequestered in adipose tissue and may be released back into systemic circulation under lipolytic conditions (e.g., fasting or exercise) (11, 12). Previous studies have shown that exercise and food deprivation can increase blood Δ^9-THC and 11-COOH-THC concentrations in rats pre-treated with THC (13, 14). Thirty-five minutes of cycling at 50% aerobic capacity (VO_{2max}) was also found to increase blood Δ^9-THC concentrations in 'regular' (i.e., ≥5 days/week) cannabis users (15). The likelihood of detecting prohibited phytocannabinoids might, therefore, be increased under certain conditions (e.g., following fasted exercise). This is significant as the WADA Code stipulates that anti-doping samples can be taken at any time – including post-competition/exercise.

The aim of the present study was therefore to investigate if supplementing with 150 mg·day of an orally consumed broad-spectrum CBD supplement for 10-weeks increased cannabinoid concentrations of CBD, Δ^9-THC and CBG in plasma and urine. In addition, we aimed to explore the influence that exercise might have in increasing metabolite concentrations following the 10-week intervention. It was hypothesised that 10-weeks of broad-spectrum supplement use would result in measurable concentrations of WADA-prohibited cannabinoids in plasma and urine biological matrices and the introduction of fasted, moderate-intensity exercise would further increase these concentrations.

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METHODS

Participants and Study Design

A randomised, double-blind, placebo-controlled parallel group trial was conducted at Liverpool John Moores University with institutional Ethics approval (21/SPS/062). Randomisation of participant group allocation was completed using an online number generator. In accordance with the funding allocation of this study, groups were distributed in an unbalanced fashion. To be eligible for participation, participants had to report no use of prescription medication and were instructed to avoid consuming any cannabis-related products for 3-months before, and throughout the experimental period. In addition, no participants were competing in any sports that were signatories to the WADA Code at the time of testing. Finally, all participants confirmed they were not taking part in any other study and agreed to use a reliable form of contraception during and 3-months following participation to mitigate the risk of fetal cannabinoid exposure (Wu, 2011). Participants were instructed to continue their typical exercise regime throughout the experimental period and instructed to report any changes or injuries to the lead researcher. Participants were invited to provide written informed consent and thereafter completed health screening and readiness to exercise questionnaires, and if deemed fit to participate, participants were included in the study, which was completed in accordance with the Declaration of Helsinki.

Protocol

Participants arrived at ~8:00am for visit 1 (Figure 1) following an overnight fast. Here, they received an information sheet and completed the Physical Activity Readiness Questionnaire (PAR-Q) form and a health screening questionnaire. Thereafter, stature was measured using a free-

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standing portable stadiometer (Seca, Hamburg, Germany), and body mass (kg) was assessed using (Seca mBCA 515; Seca, Hamburg, Germany). Thereafter, participants provided samples of venous blood and urine. Peak oxygen consumption was then assessed via a cycle ergometer-based incremental step test.

Finally, participants were familiarised with supplement ingestion and given one 15 mL bottle of their assigned treatment (containing a total of 5350 mg CBD or placebo) (detailed within the treatments section below) to take home. During visit 2 (following 5 weeks of supplementation), participants were given a second 15 mL bottle of their assigned treatment. The treatments were supplied in two separate batches as a safety measure given that the highest acute dose of CBD to have demonstrated tolerability is 6000 mg (16).

Following 10-weeks of supplementation, participants attended the laboratory at ~8:00am in a fasted state, but were encouraged to arrive well-hydrated and having taken their supplement 60-mins before arrival. This visit began with participants providing measures of body mass (kg) in addition to blood and urine samples. Individuals then performed a 90-minute bout of moderate-intensity exercise followed by a 5-min active recovery. Following the recovery, participants provided additional blood and urine samples.

Blood and urine sampling

Blood was collected into 10 mL pre-treated ethylenediaminetetraacetic acid (EDTA) and serum separator tube (SST) vacutainers (BD Biosciences, NJ, USA). Following collection, samples were inverted 10-times before centrifugation (Eppendorf 5810 R Centrifuge, Eppendorf, Hamburg,

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Germany) for 15-minutes at 2500 RCF and 4°C. Aliquots (1 mL) of supernatant were then stored at -80°C. Participants collected mid-flow urine samples into pre-labelled, sterile cups. Urine samples were rendered acellular via 15-min centrifugation at 2500 RCF and ~16°C. Thereafter, 1 mL aliquots were stored at -80°C.

Assessment of Peak Oxygen Consumption

VO_{2peak} was assessed using a graded exercise test. All exercise was performed on a cycle ergometer (Lode Corival, Groningen, The Netherlands). Before exercise began, participants were fitted with a chest-strap heart rate monitor (Polar H10; Polar, Kempele, Finland) and an appropriately fitting Hans-Rudolph 7450 facemask (Cortex Medical, Leipzig, Germany), which was connected to a pre-calibrated Cortex 3B metalyzer (Cortex Medical, Leipzig, Germany) for breath-by-breath gas analysis throughout the exercise protocol. Following a submaximal 5-minute warm-up (25 Watts [W] for females; 60 W for males) participants began the step protocol which increased in intensity by 35 W every 3 minutes until participants reached task-failure. In the final 30-seconds of each "step" researchers asked participants to provide a rating of perceived exertion (RPE; 6-20 Borg scale (17)). At the same time as RPE rating, a finger prick capillary blood sample was taken, which was analysed using a Biosen C-line Glucose and Lactate analyser (EKF Diagnostics, Cardiff, UK) for blood lactate concentration assessment.

Treatments

The active and placebo products used in this study were both 100% vegan certified and third-party (Spring Creek Labs, Utah, USA; see Supplemental Digital Content) laboratory tested for cannabinoids, residual solvents, pesticides, heavy metals and microbials. The product was a

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commercially available, off-the-shelf, oral formulation of CBD. The product came in two dropper bottles containing 15 mL of oil. Independent testing (Supplemental Content) evidenced a CBD concentration of 395.13 mg/g (43.184%), CBG at 8.83 mg/g (0.5955%), CBDV (not reported), with low Δ^9-THC (< 0.00025 % with 0.00025% the limit of detection). The placebo group received a 2 x 15 mL, visually identical dropper bottles of MCT oil. Participants were familiarised with supplement ingestion, which comprised self-administrating 9-drops (16.67 mg/drop) of their treatment sublingually, waiting for 30-seconds and then swallowing. This dose was elected as it was similar to that used in other studies involving CBD and exercise (Cochrane-Snyman et al., 2021; McCartney et al., 2024), in addition to it being considerably below the maximum reported safe dose (Taylor et al., 2018).

Participants were instructed to consume their supplements at the same time (~8 am +/- 1 hour) each day for 10 weeks, and to store their supplements in a cool, dry place out of the reach of children. Participants were instructed to consume their supplement without food so to avoid any nutrition-related variability in CBD absorption. In addition, participants were required to inform the lead researcher if they failed to consume a dose and were told explicitly not to double dose the next day. There were no reported missed or double doses of participants who were within the final sample. The final dose of the participants' respective supplement was consumed 60-mins before arrival at the laboratory on the final day of testing.

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Blinding

Participants were randomised into two groups, experimental (CBD) or control (PLA). Randomisation was completed using a commercially available random number generator by a staff member who was independent of the study.

Moderate Intensity Exercise Protocol

Before exercise commenced, participants' body mass was assessed as per visit 1. Thereafter, a blood draw was taken via venepuncture of the antecubital vein. Following which, participants provided a mid-stream urine sample. Exercise commenced with a 5-minute warm-up (as per visit 1). Participants then completed a 90-minute bout of moderate-intensity exercise at ~55% of the final wattage achieved during the initial $\dot{V}O_{2peak}$ protocol. Breath-by-breath indirect calorimetry was performed throughout; with average heart rate (beats×min$^{-1)}$, and respiratory exchange ratio (RER) calculated over 10-minute periods throughout exercise. Perceived exertion (RPE [Borg Scale; 6-20]) and blood lactate (BLa) concentrations (mmol×L^{1}) were also collected in the final 30-seconds of each 10-minute period during exercise. Upon completion of the exercise protocol participants repeated the pre-exercise measures (body mass, urine sample and blood draws), which were all completed within 30-minutes of exercise cessation.

Biological Samples

Plasma Cannabinoids. Plasma cannabinoid concentrations were measured using ultra-high performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) and previously validated methods (Kevin et al., 2021). Briefly, samples underwent protein precipitation followed by supported liquid extraction. Gradient mode LC-MS/MS was then used to separate and quantify

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the target analytes. All samples were analysed in triplicate, with those derived from the same participant analysed on the same plate.

Urinary Cannabinoids

Reagents. Cannabinoid standards and deuterated internal standards: THC, THC-d3, CBD, CBD-d3, CBD, CBDV, 11-OH-THC, 11-OH-THC-d3, THC-COOH, THC-COOH-d9, THC-COOH-Glucuronide, and THC-COOH-Glucuronide-d3 were obtained from Cerilliant (Round Rock, TX, USA). 6-OH-CBD, 7-OH-CBD, 7-COOH-CBD, 6-OH-CBD-d9, 7-OH-CBD-d10, 7COOH-CBD-d10 were obtained from BDG Synthesis (Wellington, New Zealand). LCMS grade methanol, acetonitrile, n-hexane, aqueous ammonia, anhydrous sodium acetate, glacial acetic acid and formic acid were acquired from Fisher Scientific (Melbourne, VIC, Australia); and Kura Biotech, BG100 red abalone b-glucuronidase from PM Separations (Capalaba, QLD, Australia). All chemicals and solvents were at least HPLC or LCMS grade, respectively.

Sample Preparation

Urine samples were first thawed and 50 µL from each collected sample was transferred into a glass-lined 96-well plate. Thereafter, the calibrator and quality control (QC) samples were prepared by spiking the cannabinoid-free "blank", to reach a maximum volume of 5 µL. All samples were then spiked with internal standard solutions and 5 µL MeOH was added to all study samples.

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Hydrolysis

The sample pH was first adjusted to 4.5-5.0 by adding 200 µL of 0.1M acetate buffer (pH 4.4), and this was briefly vortexed. Thereafter 50 µL of β-glucuronidase (5000U; BG100, Kura Biotech) was added and vortexed for 2 minutes. This solution was then incubated at 65°C for 30 min. The sample pH was finally adjusted to 2.5-3.0 via the addition of 10 µL formic acid, and briefly vortexed.

Supported Liquid Extraction (SLE)

Samples were immediately transferred onto a 400 µL SLE 96-well- plate (Biotage ISOLUTE® SLE+). A brief nitrogen push was then performed, and the solution was then left to equilibrate for 5-min at room temperature. Analytes were then transferred into a fresh glass-lines 96-well-plate via a two-step method. First, 700 µL/sample Dichloromethane was added and allowed to flow under gravity for 5 min followed by a nitrogen gas push of 0.5 bar for 30 seconds. Methyl tert-butyl ether (900 µL/sample) was then added and again allowed to flow under gravity for 5 min followed by a nitrogen gas push of 0.5 bar for 120 seconds. A final push of 5 bars of nitrogen was then applied for 120 seconds. This eluate was then evaporated under a gentle stream of nitrogen for 45 mins. Samples were then reconstituted in their initial LCMS-MS mobile phase condition of 60% Mobile phase B (MeOH) and 40% Mobile phase A (0.1% Forminc Acid [FA] water) in two-steps, to a total volume of 100 µL, as follows: To a blank well containing internal standard only, 60 µL of MeOH containing 11-COOH-THC-Glucuronide and 11-COOH-THC-Glucuronide-d3 and 40 µL of 0.1% Formic Acid in water was added to reach 500 $ng \cdot ml^{-1}$of 11-COOH-THC-Glucuronide and 50 $ng \cdot ml^{-1}$ final concentration. To all proceeding wells 60µl MeOH

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was added, followed by 40 µL 0.1% FA water, and vortexed for 5 minutes, before the plate was sealed for immediate analysis.

Statistical Analysis

The plasma and urine cannabinoid concentrations of the intervention (i.e., CBD) group were analysed using Two-way repeated measures ANOVA that included time (baseline, pre-exercise, post-exercise). The PLA groups plasma and urine data were not formally statistically analysed as these were only ever intended to verify abstinence from external cannabis/CBD use. Statistical significance was defines as $p < 0.05$. Data are presented as means ± SD with 95% confidence intervals (CI). Statistical analysis was performed in IBM SPSS Statistics for Windows (Version 30.0.0.0; Armonk, New York, USA) and figures were prepared in GraphPad Prism V.9 for Macintosh (GraphPad Software, San Diego, California, USA).

Sample Size

For a moderate effect size ($f = 0.25$), for one group and three measurements, and for an alpha level of 0.05 and power of 0.8, a sample size of 28 participants was calculated (Faul et al., 2007). To account for dropout, 31 participants were recruited in the CBD group, with 5 participants in the PLA group. The unbalanced design was chosen to maximise the analysis of cannabinoids and their metabolites following supplementation and exercise. This is common when assessing the presence, distribution, metabolism or excretion of a compound, including its metabolites because only those exposed to the active compound will exhibit the measurable analytes making it both ethically and scientifically justifiable.

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RESULTS

Participant Characteristics

Thirty-six healthy and physically active males (n = 17) and females (n = 19) (mean ± SD: age: 27 ± 6 years; stature: 171.9 ± 10.4 cm; body mass: 75.69 ± 16.82 kg; body fat: 25.00 ± 6.88 %; VO_{2peak}: 39.2 ± 8 ml/kg/min) were recruited between 12.10.2021 and 04.11.2023 (Table 1). All 36 participants were randomised, completed the 10-week supplementation period and were included in the final analytical sample. The final analytical sample included the CBD group only (n = 31; 15 males, 16 females).

Blinding

Upon completion of the study, participants were asked which group (CBD/PLA) they believed they were in. Twelve of the 36 (33%) participants correctly identified their assigned treatment condition.

Cannabinoid Responses to Supplementation

No phytocannabinoids were present in either urine or plasma at baseline in either the CBD or PLA groups. Therefore, statistical analyses reported herein are comparisons of cannabinoid metabolites following 10-weeks of supplementation (pre-exercise) and post-exercise. In addition, 6-OH-CBD was not detectable in any plasma sample. Δ^9-THC, 11-OH-THC and 11-COOH-THC were not detected in any urine or plasma samples (i.e., from either group) at any time during the study. All other cannabinoids and their metabolites were observed in plasma or urine in the CBD group, as described in the following sections. All data can be found in (Table 2,3) and (Figure 2, 3).

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Plasma Cannabinoids

There was a significant effect of time on CBD ($F_{4.501}$, P = 0.043) indicating both groups showed a significant increase in this metabolite following exercise (Figure 2A). There was no significant effect of sex ($F_{.000}$, P = 0.983). There was a significant effect of time on 7-OH-CBD ($F_{11.919}$, P = 0.002) indicating both groups showed a significant increase in this metabolite following exercise (Figure 2B). There was no significant effect of sex ($F_{.314}$, P = 0.579). There was a significant effect of time on 7-OH-CBD ($F_{4.542}$, P = 0.042) indicating both groups showed a significant increase in this metabolite following exercise (Figure 2C). There was no significant effect of sex ($F_{.244}$, P = 0.625).

Urinary Cannabinoids

There was a significant effect of time on CBD ($F_{4.501}$, P = 0.043) indicating both groups showed a significant increase in this metabolite following exercise (Figure 3A). There was no significant effect of sex ($F_{.000}$, P = 0.983). There was a significant effect of time on CBDV ($F_{5.008}$, P = 0.033) indicating both groups showed a significant increase in this metabolite following exercise (Figure 3B). There was no significant effect of sex ($F_{.198}$, P = 0.659). There was a significant effect of time on CBG ($F_{5.754}$, P = 0.023) indicating both groups showed a significant increase in this metabolite following exercise (Figure 3C). There was no significant effect of sex ($F_{.339}$, P = 0.565). There was a significant effect of time on 6-OH-CBD ($F_{8.626}$, P = 0.006) indicating both groups showed a significant increase in this metabolite following exercise (Figure 3D). There was no significant effect of sex ($F_{.002}$, P = 0.966). There was no significant effect of time on 7-COOH-CBD ($F_{2.458}$, P = 0.128) indicating neither group showed a significant increase in this metabolite following exercise (Figure 3E). There was no significant effect of sex ($F_{.154}$, P = 0.698).

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There was however a significant effect of time on 7-OH-CBD ($F_{7.510}$, P = 0.010) indicating both groups showed a significant increase in this metabolite following exercise (Figure 3F). There was no significant effect of sex ($F_{.239}$, P = 0.629).

DISCUSSION

The present study measured plasma and urinary concentrations of three cannabinoids and three their metabolites following 10-weeks of supplementation with 150 mg per day of a broad-spectrum CBD product. The secondary aim of this investigation was to identify whether cannabinoid concentrations might be magnified by a combination of fasting and moderate-intensity exercise on the assumption that cannabinoids may accumulate in adipose tissue and be mobilised by the lipolytic stimulus of fasted exercise. We report that 10-weeks of daily supplementation with 150 mg per day of an orally-consumed broad-spectrum CBD product significantly increased concentrations of various cannabinoid metabolite concentrations in plasma and urine, excluding those associated with Δ^9-THC. We also demonstrate that fasted, moderate-intensity exercise significantly increases urinary concentrations of WADA prohibited cannabinoids including CBG and CBDV. Finally, we demonstrate that there was no effect of biological sex on any of the cannabinoids or metabolites measured in this study.

This is the first study to illustrate the significant anti-doping risks for athletes supplementing with an off-the-shelf CBD product. Our data indicate conclusively that if an athlete were to consume a broad-spectrum CBD supplement, they are at risk of an adverse analytical finding (18). Importantly, the risk is not due to Δ^9-THC (at least, in the context of this broad-spectrum CBD product), but rather to "minor" cannabinoids such as CBG and CBDV. These risks

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are elevated significantly by exercise as we show increases in urinary concentrations of these prohibited minor cannabinoids. Collectively, our data highlight the potential anti-doping risks for athletes using broad-spectrum CBD products and a need for clarity and education from athlete and WADA perspectives, respectively.

The pharmacokinetics and sequestration of cannabinoids have been a topic of debate for some time (19, 20), yet few studies have characterised the accumulation of cannabinoids over an extended period in healthy individuals. Upon ingestion, CBD undergoes phase I metabolism, which primarily occurs via hydroxylation and subsequent carboxylation by liver cytochrome P450 enzymes (21). These processes give rise to several metabolites, namely, 6-OH-CBD, 7-OH-CBD, 7-COOH-CBD (22). Our data are consistent with previous studies, that show that 7-COOH-CBD can be observed in significantly greater concentrations than CBD, 6-OH-CBD and 7-OH-CBD, which might make 7-COOH-CBD metabolite a more appropriate method of monitoring peak cannabinoid concentrations (23).

Regarding sequestration, little is known regarding the effects of exercise and/or fasting on cannabinoid concentrations. In rats, lipolysis has been suggested to stimulate the release of cannabinoids from adipose tissue. The authors speculated that the animals in their study could re-intoxicate with Δ^9-THC (13). In humans, it has been reported that exercise increased plasma concentrations of Δ^9-THC following 35-minutes of aerobic exercise in some non-fasted regular cannabis users. However, fasting for 12-hours (independent of moderate-intensity exercise) did not significantly increase plasma Δ^9-THC or THC-COOH (15). We postulate that the lipolytic effect of moderate-intensity exercise can significantly affect cannabinoid release from adipose

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tissue. Our data support the conclusions of these studies and indicate that participants in the experimental group had significantly greater post-exercise urinary concentrations of various cannabinoids and their metabolites compared to those evidenced before exercise commenced. We also reasoned that female participants may show a different response to supplementation due to generally higher body fat percentage and potential for greater cannabinoid sequestration, coupled with sex differences in CYP enzyme activity. CBD is metabolized primarily by CYP3A4, CYP2C19, and CYP2C9 (Zendulka et al., 2016; Ujváry & Hanuš, 2016). CYP3A4 activity is typically higher in females (Wolbold et al., 2003), which may lead to faster metabolism and increased production of certain CBD metabolites. Other enzymes such as CYP2C19 and CYP2C9 may exhibit more variable or male-biased activity, contributing to sex-specific differences in metabolite profiles (Anderson, 2005; Zanger & Schwab, 2013). However, we observed no sex differences in any of the detectable cannabinoids or their metabolites in our study.

Perhaps most notable in our investigation were the findings associated with minor cannabinoids, which are considered non-threshold substances by WADA. Despite their relatively low concentrations within the formulation, CBG (8.83 $mg.g^{-1}$; 0.5955%) and CBDV (not-reported on Certificate of Analysis [CoA; Supplementary Material]) compared to CBD (395.13 $mg.g^{-1}$; 43.1840%), both were present in detectable concentrations in urine. Specifically, urinary concentrations of CBG were present in 42% of pre-exercise samples (18 ± 44 $ng.ml^{-1}$) and 74% of post-exercise samples (54 ± 77 $ng.ml^{-1}$). Similarly, urinary concentrations of CBDV were present in 68% of pre-exercise samples (26 ± 55 $ng.ml^{-1}$) and 84% of post-exercise samples (10 ± 63 $ng.ml^{-1}$).

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It is also important to consider the half-lives of orally consumed cannabinoids. For example, recent reports have suggested that the half-life of CBD in humans might be >134 hours (24). Similarly, it has been reported that Δ^9-THC has a dynamic pharmacokinetic profile, with a terminal half-life of ~72-96 hours (25). However, our data indicate that given Δ^9-THC within the supplement used was <0.00025%, neither Δ^9-THC nor its metabolites were detectable in the present study. There is, however, a paucity of data on the pharmacokinetics of CBG and/or CBDV in humans. Despite the detection of these metabolites in our samples, we did not directly assess half-life and therefore future studies could look to address this.

Limitations

We acknowledge the limitations of the present study. Firstly, the CBD supplement provided was self-administered. Though participants were familiarised with correct supplement ingestion, adherence was not monitored, although our data indicate that all participants in the CBD group had increased CBD concentrations after 10-weeks of supplementation. Nevertheless, we present substantial between-subject metabolite concentration variability in line with previous reports (26). While our participants were instructed to consume their respective supplements 60 mins before presenting at the laboratory, we show marked variability in pre-exercise cannabinoid concentrations. In addition, despite the relatively poor oral bioavailability of cannabinoids, reported to be as low 6% in some cases (27, 28) – we evidence fair detection in all of those examined. However, other variables known to influence cannabinoid pharmacokinetics include the co-administration of food. For example, previous studies have evidenced that high-fat meals can influence CBD pharmacokinetics (29). We acknowledge that overnight fasting may have limited peak plasma concentrations in our study, however this design was necessary to maximise lipolysis

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during exercise (30). While the number of participants is greater than some other investigations of this kind, a larger pool of participants would be beneficial in future studies. We also hypothesise that the increase in cannabinoids in plasma and urine following fasted exercise may be due to liberation of these metabolites from adipose tissue. However, it must be acknowledged that we cannot confirm this from the current study design. It could be argued that changes in plasma volume following exercise explain the increase concentration of cannabinoids in plasma. However, males in our study had an average loss of body mass of 1.2kg pre to post exercise, whereas females experienced a 260g loss following exercise, yet there were no sex-differences at baseline or following exercise in any cannabinoid or associated metabolite. Moreover, the magnitude of change in the cannabinoids we detected following exercise are beyond what would reasonably occur due to such changes in plasma volume. To test our hypothesis, future studies could utilise inhibitors of adipose tissue lipolysis, such as nicotinic acid, to confirm whether the same exercise-induced increases in plasma and urinary cannabinoids and associated metabolites still occur when lipolysis is significantly blunted. Finally, we used only a single brand of broad-spectrum CBD. Therefore, results should be interpreted with caution, and further independent analysis of other broad-spectrum CBD products is advisable.

CONCLUSIONS

We present the first data to demonstrate that consuming a broad-spectrum CBD supplement containing trace amounts of WADA-prohibited cannabinoids results in detectable increases in their presence in urine. In addition, our data provide a novel insight that cannabinoid concentrations are increased following acute moderate-intensity exercise. Neither the presence of cannabinoids in these biological matrices nor the response to exercise was affected by biological sex in our study.

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These data add further support to our previous recommendations (Gillham et al., 2024) that athletes should abstain from the use of CBD supplements until there is either **a**) batch-tested CBD isolate available confirming the complete absence of all minor cannabinoids or **b**) WADA policy regarding cannabinoids is changed and minor cannabinoids are considered threshold substances like Δ^9-THC, or specific cannabinoids are named as prohibited rather than naming the CBD as the only cannabinoid not prohibited.

Acknowledgements

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FIGURE LEGENDS

Figure 1: Schematic representation of study procedures. Participants visited the laboratory on three occasions over a 10-week supplementation period. Blood and urinary cannabinoid concentrations were measured at visit 1 (baseline) and visit 3 (pre- and post-exercise).

Figure 2: Plasma concentrations of A) CBD; B) 7-COOH-CBD; C) 7-OH-CBD at baseline, pre-exercise (10-weeks) and post-exercise (10-weeks) in the CBD group for males and females (n = 31). * Denotes a significant difference between pre - and post-exercise ($P < 0.05$). Values are expressed as mean ± SD.

Figure 3: Urinary concentrations of A) CBD; B) CBDV; C) CBDG; D) 6-OH-CBD; E) 7-COOH-CBD; F) 7-OH-CBD at baseline, pre-exercise (10-weeks) and post-exercise (10-weeks) for males and females in the CBD group (n = 31). * Denotes a significant difference between pre - and post-exercise ($P < 0.05$). Values are expressed as mean ± SD.

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SUPPLEMENTAL DIGITAL CONTENT

SDC 1: CoA – 40% Oil.pdf

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Figure 1

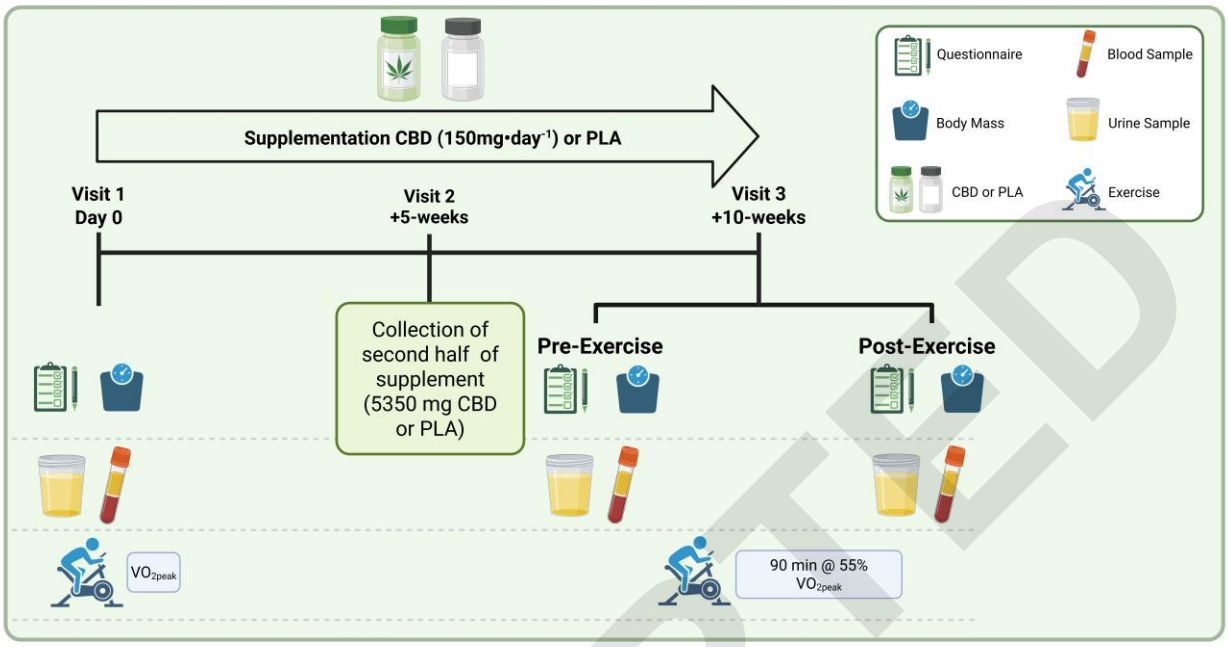

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Figure 2

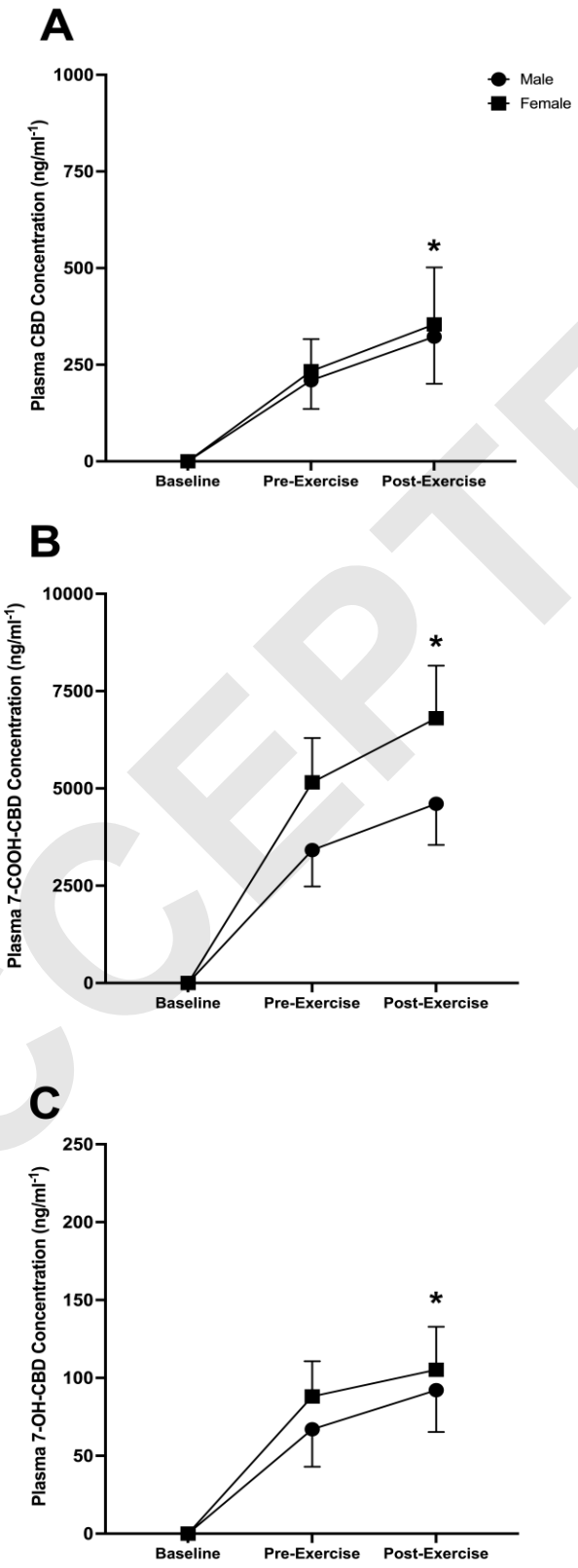

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Figure 3

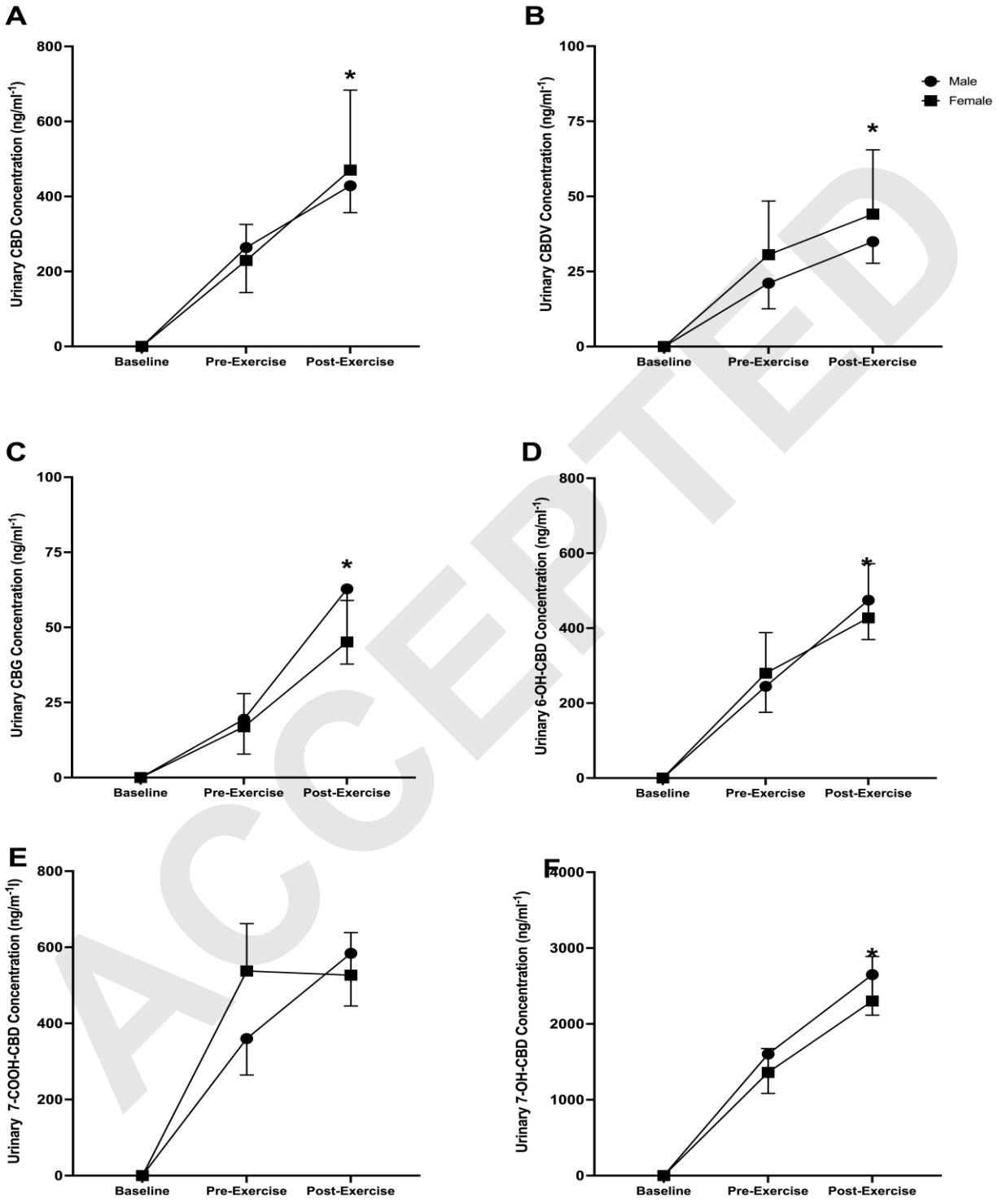

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Table 1. Participants (n=36) physical responses to the moderate intensity exercise (90-minutes at 55% VO_{2peak}) bout.

Variable	Group	CBD	PLA
HR ($b \cdot min^{-1}$)	Male	132 ± 18 (95% CI = 129.48-135.32)	125 ± 21 (95% CI = 114.96 – 134.54)
	Female	136 ± 24 (95% CI = 132.40-1439.77)	125 ± 21 (95% CI = 126.88 – 143.95)
BLa ($mmol \cdot l^{-1}$)	Male	3.20 ± 1.87 (95% CI = 2.77 – 3.63)	3.88 ± 2.80 (95% CI = 1.88 – 5.88)
	Female	2.30 ± 1.25 (95% CI = 2.03 – 2.57)	2.81 ± 0.35 (95% CI = 2.61 – 3.02)
RER	Male	0.92 ± 0.05 (95% CI = 0.91-0.93)	0.90 ± 0.05 (95% CI = 0.88- 0.92)
	Female	0.91 ± 0.05 (95% CI = 0.90-0.91)	0.92 ± 0.05 (95% CI = 0.90-0.94)
VO2 ($L.min^{-1}$)	Male	2.06 ± 0.52 (95% CI = 1.97 – 2.14)	2.36 ± 0.47 95% CI = 2.14 – 2.58)
	Female	1.38 ± 0.35 95% CI = 1.32 – 1.43)	1.12 ± 0.21 (95% CI = 1.01-1.20)
RPE (AU; 6-20)	Male	12.26 ± 2.52 (95% CI = 11.86 – 12.67)	11.85 ± 2.50 (95% CI = 10.68 – 13.02)
	Female	11.93 ± 2.39 (95% CI = 11.55 – 12.30)	15.53 ± 2.91 (95% CI = 12.45 – 1.62)
Pre Exercise Body Mass (kg)	Male	87.66 ± 16.10 (95% CI = 78.12 – 97.19)	86.1 ± 13.55 (95% CI =-86.07-258.27)
	Female	65.06 ± 6.84 (95% CI = 61.14 – 68.98)	67.82 ± 6.29 (95% CI = 48.69 – 86.94)
Post Exercise Body Mass (kg)	Male	86.37 ± 16.82 (95% CI = 78.12 – 97.19)	85.63 ± 16.63 (95% CI =-87.5 – 258.75)
	Female	67.35 ± 6.15 (95% CI = 48.64 – 86.06)	67.35 ± 6.15 (95% CI = 48.64 – 86.06)

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Table 2. Plasma concentrations of CBD (n=31) and PLA (n=5) groups across time.

Metabolite	Condition	Male			Female		
		Baseline	Pre-Exercise	Post-Exercise	Baseline	Pre-Exercise	Post-Exercise
7-COOH-CBD	CBD	0.00 ± 0.00	3491.71 ± 3502.07 (95% CI = 1408.39 – 5423.29)	4781.86 ± 43961.85 (95% CI = 2333.45 – 6875.46)	0.00 ± 0.00	5154.71± 45410.99 (95% CI = 2727.18 – 7582.25)	6804.23 ± 5227.64 (95% CI = 3927.26 – 9681.20)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
7-OH-CBD	CBD	0.00 ± 0.00	66.82 ± 89.94 (95% CI = 15.42 – 118.53)	97.11 ± 100.33 (95% CI = 34.63 – 149.66)	0.00 ± 0.00	88.06 ± 87.80 (95% CI =39.73 – 136.38)	105.19 ± 107.03 (95% CI = 46.29 – 164.09)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
CBD	CBD	0.00 ± 0.00	197.73 ± 275.95 (95% CI = 51.38 – 149.66)	333.13 ± 454.16 (95% CI = 6195 – 582.62)	0.00 ± 0.00	233.39 ± 321.34 (95% CI = 56.54- 410.23)	354.50 ± 570.20 (95% CI = 40.70 – 668.30)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

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Table 3. Urinary concentrations of CBD (n=31) and PLA (n=5) groups across time.

Metabolite	Condition	Male			Female		
		Baseline	Pre-Exercise	Post-Exercise	Baseline	Pre-Exercise	Post-Exercise
6-OH-CBD	CBD	0.00 ± 0.00	239.87 ± 26 (95% CI = 95.9 – 393.83)	498.53 ± 393.25 (95% CI =249.35 – 700.18)	0.00 ± 0.00	279.64 ± 419.90 (95% CI = 48.55 – 510.72)	427.44 ± 559.66 (95% CI = 119.44 – 735.45)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
7-COOH-CBD	CBD	0.00 ± 0.00	369.40 ± 357.56 (95% CI = 155.03 – 564.95)	611.70 ± 517.85 (95% CI = 287.10 – 880.78)	0.00 ± 0.00	537.66 ± 481.82 (95% CI = 272.49 – 802.82)	526.83 ± 433.68 (95% CI = 288.16 – 765.49)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
7-OH-CBD	CBD	0.00 ± 0.00	1661.26 ± 1941.28 (95% CI = 490.58 – 2716.14)	2787.89 ± 2003.43 (95% CI = 1501.94-3798.74)	0.00 ± 0.00	1360.56 ± 1218.82 (95% CI = 6893.80 – 2031.32)	2302.39 ± 2270.23 (95% CI = 1053.00-3551.78)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
CBD	CBD	0.00 ± 0.00	272.34 ± 449.77 (95% CI = 6.07 – 521.69)	454.58 ± 268.91 (95% CI = 274.33 – 582.62)	0.00 ± 0.00	229.41 ± 371.75 (95% CI = 24.82 – 434.00)	470.26 ± 826.11 (95% CI = 15.62 - 924.90)

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	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
CBG	CBD	0.00 ± 0.00	20.81 ± 43.48 (95% CI =-5.5-44.35)	67.29 ± 93.64 (95% CI = 9.13 – 116.48)	0.00 ± 0.00	16.91 ± 42.76 (95% CI =-6.62 – 40.44)	45.14 ± 53.49 (95% CI = 15.7 – 74.57)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
CBDV	CBD	0.00 ± 0.00	21.27 ± 31.95 (95% CI = 2.78 – 39.41)	36.80 ± 26.94 (95% CI = 19.51 – 50.39)	0.00 ± 0.00	30.58 ± 69.12 (95% CI =-7.46 – 68.61)	44.08 ± 82.99 (95% CI =-1.6 – 89.75)
	PLA	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

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Certificate of Analysis

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Sample: 2010TSF0214.10383
Strain: 40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml
Batch #: 2010059; Lot #: 2010059;
Sample Received: 10/26/2020; Report Generated: 10/30/2020

40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml

Ingestible, Tincture, Other
Harvest Process Lot: ; METRC Batch: ; METRC Sample:

The photo on this report is of a sample collected by the lab and may vary from the final packaging

<LOD	4,148.87 mg/unit
Total Potential THC	Total Potential CBD

Cannabinoids

Analyte	LOD	LOQ	Result	Result
	mg/unit	mg/unit	mg/unit	mg/g
THCa	1.05	1.05	ND	ND
Δ9-THC	1.05	1.05	ND	ND
Δ8-THC	1.05	1.05	ND	ND
CBD	1.05	1.05	4148.87	395.13
CBDa	1.05	1.05	ND	ND
CBC	1.05	1.05	ND	ND
CBG	1.05	1.05	92.72	8.83
CBN	1.05	1.05	ND	ND
THCV	1.05	1.05	ND	ND
CBGa	1.05	1.05	ND	ND
Total			**4241.58**	**403.96**

1 Unit = , 10.5g

Test performed via HPLC-UV. Total Potential THC and CBD: Liquid chromatography occurs at room temperature and does not decarboxylate any cannabinoids, thereby yielding separate values for THCa, THC, CBDa and CBD, which are then combined to derive the Total Potential THC and Total Potential CBD result using the following formulae:
Total Potential THC = THCa * 0.877 + Δ9-THC+ Δ8-THC
Total Potential CBD = CBDa * 0.877 + CBD
LOQ = Limit of Quantitation; Cannabinoids for flower and trim reported as received. LOQ has been set to the Limit of Detection (LOD) / Method Detection Limit (MDL) for all endpoints.

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Dr. Jeff Angermann
Scientific Director

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Sample: 2010TSF0214.10383
Strain: 40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml
Batch #: 2010059; Lot #: 2010059;
Sample Received: 10/26/2020; Report Generated: 10/30/2020

40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml

Ingestible, Tincture, Other
Harvest Process Lot: ; METRC Batch: ; METRC Sample:

Residual Solvents

Pass

Analyte	Result	LOQ	Limit	Status
	PPM	PPM	PPM	
Butanes	<LOQ	5	500	Pass
Ethanol	<LOQ	5		Tested
Heptanes	<LOQ	5	500	Pass
Isobutane	<LOQ	5	500	Pass
Pentane	<LOQ	5		Tested
Propane	<LOQ	5	500	Pass

ND = Non Detect; LOQ = Limit of Quantitation; Test performed via Headspace GC-FID. LOQ has been set to the Limit of Detection (LOD) / Method Detection Limit (MDL) for all endpoints.

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Ingestible, Tincture, Other
Harvest Process Lot: ; METRC Batch: ; METRC Sample:

Pesticides — Pass

Analyte	LOQ	Limit	Result	Status
	PPM	PPM	PPM	
Abamectin	0.100	0.200	<LOQ	Pass
Acequinocyl	2.000	4.000	<LOQ	Pass
Beta-Cyfluthrin	0.200	2.000	<LOQ	Pass
Bifenazate	0.100	0.400	<LOQ	Pass
Bifenthrin	0.100	0.100	<LOQ	Pass
Cyfluthrin	0.200	2.000	<LOQ	Pass
Cypermethrin	0.200	1.000	<LOQ	Pass
Daminozide	0.100	0.800	<LOQ	Pass
Dimethomorph	0.100	2.000	<LOQ	Pass
Etoxazole	0.100	0.400	<LOQ	Pass
Fenhexamid	0.100	1.000	<LOQ	Pass
Flonicamid	0.100	1.000	<LOQ	Pass
Fludioxonil	0.100	0.500	<LOQ	Pass
Imidacloprid	0.100	0.500	<LOQ	Pass
Myclobutanil	0.100	0.400	<LOQ	Pass
Paclobutrazol	0.100	0.400	<LOQ	Pass
Piperonyl Butoxide	0.100	3.000	<LOQ	Pass
Plant Growth Regulators			<LOQ	Tested
Pyrethrins	0.500	2.000	<LOQ	Pass
Quintozene	0.100	0.800	<LOQ	Pass
Spinetoram	0.100	1.000	<LOQ	Pass
Spinosad	0.100	1.000	<LOQ	Pass
Spirotetramat	0.100	1.000	<LOQ	Pass
Thiamethoxam	0.100	0.400	<LOQ	Pass
Trifloxystrobin	0.100	1.000	<LOQ	Pass

Test performed via liquid chromatography triple quadrupole mass spectrometry and gas chromatography triple quadrupole mass spectrometry ND = Non Detect; LOQ = Limit of Quantitation; Tested Pesticides: Nevada Division of Public and Behavioral Health Monitoring List for Pest Control Substances. LOQ has been set to the Limit of Detection (LOD) / Method Detection Limit (MDL) for all endpoints.

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Dr. Jeff Angermann
Scientific Director

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All pass limits are as specified in NAC 453.A and DPBH Policies. Unless otherwise stated all quality control samples performed within specifications established by the Laboratory. This product has been tested by 374 Labs, LLC (MME# 03754325902079441647) using valid testing methodologies and a quality system as required by Nevada state law. Values reported relate only to the product tested. 374 Labs makes no claims as to the efficacy, safety or other risks associated with any detected or non-detected levels of any compounds reported herein. This Certificate shall not be reproduced except in full, without the written approval of 374 Labs. Uncertainty information is available upon request. 374 Labs complies with ISO/IEC 17025 standards. Potency performed per SOP-000001, Terpenes performed per SOP-000002, Metals performed per SOP-000003, Microbial testing performed per SOP-000004, Pesticides performed per SOP-000005. Moisture per SOP-000006. Photo is of sample collected by the lab and may vary from final packaging. LOQ has been set to the Limit of Detection (LOD) / Method Detection Limit (MDL) for all endpoints.

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Certificate of Analysis

Spring Creek Labs
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glugo2008@gmail.com
(503) 884-8056
Lic. #

Sample: 2010TSF0214.10383
Strain: 40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml
Batch #: 2010059; Lot #: 2010059;
Sample Received: 10/26/2020; Report Generated: 10/30/2020

40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml

Ingestible, Tincture, Other
Harvest Process Lot: ; METRC Batch: ; METRC Sample:

Heavy Metals — Pass

Analyte	Result	LOQ	Limit	Status
	PPM	PPM	PPM	
Arsenic	<LOQ	0.460	2.000	Pass
Cadmium	<LOQ	0.230	0.820	Pass
Lead	<LOQ	0.310	1.200	Pass
Mercury	<LOQ	0.120	0.400	Pass

Test performed via Inductively Coupled Plasma Mass Spectrometry (ICP-MS). ND = Non Detect; LOQ = Limit of Quantitation; Tested Metals: Lead, Cadmium, Mercury, Arsenic. LOQ has been set to the Limit of Detection (LOD) / Method Detection Limit (MDL) for all endpoints.

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Lic. #

Sample: 2010TSF0214.10383
Strain: 40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml
Batch #: 2010059; Lot #: 2010059;
Sample Received: 10/26/2020; Report Generated: 10/30/2020

40% - 4000mg, CBD, BS, MCT, NF, Tincture, 10ml

Ingestible, Tincture, Other
Harvest Process Lot: ; METRC Batch: ; METRC Sample:

Microbials — Pass

Analyte	Mass	LOQ	Limit	Status
	CFU/g	CFU/g	CFU/g	
Total Aerobic	<LOQ	40	100000	Pass
Aspergillus flavus	NR	1	1	NT
Aspergillus fumigatus	NR	1	1	NT
Aspergillus niger	NR	1	1	NT
Aspergillus terreus	NR	1	1	NT
Total Enterobacteriaceae	<LOQ	40	1000	Pass
Total Coliforms	<LOQ	40	1000	Pass
Pathogenic E. Coli	<LOQ		1	Pass
Salmonella	<LOQ		1	Pass
Total Yeast & Mold	<LOQ	40	10000	Pass

The photo on this report is of a sample collected by the lab and may vary from the final packaging

Test performed via Hardy chromogenic dry and agar plates. ND = Non Detect; LOQ = Limit of Quantitation; Tested Microbial Endpoints: Total Viable Aerobic Bacteria, Total Yeast and Mold, Total Coliforms, Enterobacteriaceae, Aspergillus (flavus, fumigatus, niger and terreus), E.coli pathogenic strains, and Salmonella spp. Microbial testing performed per SOP-000004 Method. LOQ has been set to the Limit of Detection (LOD) / Method Detection Limit (MDL) for all endpoints.

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